HOSPITAL MANAGEMENT SYSTEM

UNIT TEST REPORT

Software Engineering Project · C++17 / CTest automated test execution · 6 November 2026

PRIMARY RESULT	51 of 51 registered tests passed | 100% pass rate

51 / 51	0	0	31.47 s
Tests passed	Failed	Skipped	Total test time

1. Purpose and scope

This report records the registered automated test suite and its execution outcome for the C++ Hospital Management System. The suite covers patient and doctor management, appointment workflows, CSV persistence, user-interface behaviour, robustness, performance, privacy, security, architectural isolation, and offline operation. The inventory below preserves the registered test names and concise behaviour descriptions from the project test inventory.

2. Test environment and build configuration

Item	Recorded value
Language / standard	C++17
Build system / generator	CMake 3.20+ / Ninja
Compiler	MSYS2 CLANG64 · Clang 23.1.5
Build configuration	Debug
Test runner	CTest
Registered / executed	51 / 51
Execution outcome	51 passed · 0 failed · 0 skipped
Recorded duration	31.47 seconds
Source revision	542b90be8adaddf13f7ce900340973eb4c9f5a56

3. Aggregate execution result

Execution metric	Result
Registered tests	51
Executed tests	51
Passed	51
Failed	0
Skipped	0
Pass rate	100%
Total duration	31.47 seconds
JUnit output	build-llvm/ctest-results.xml

4. Test coverage by functional area

Area	Representative registered checks
Patient management	Registration, field validation, duplicate IDs, search, updates, identity protection, persistence
Doctor management	Creation, validation, duplicate IDs, search, listing, updates, identifier protection
Appointments	Booking, unknown entities, availability, conflicts, cancellation, ordering, filtering
Persistence / CSV	Startup loading, missing/malformed files, write failures, duplicate/orphan records, readable CSV
UI and robustness	Menu navigation, invalid selections, operation messages, clean exit, startup resilience
Security and privacy	Input validation, formula-injection guard, over-length input, data minimization, safe errors, directory restriction
Non-functional	1,000-record performance scenario, data retention, menu depth, class isolation, offline operation

UNIT TEST REPORT — TEST INVENTORY

Registered CTest cases and recorded execution status · 6 November 2026

5. Complete registered test case inventory

Formal IDs are shown where supported by the available test documentation. 'CTest-only' means the registered CTest target is the reference; no separate formal ID is established.

No.	ID / ref.	Registered CTest name	Objective / behaviour checked	Result
1	TC-PAT-02	DuplicatePatientId	Rejects registration using an existing patient ID.	PASS
2	TC-PAT-03	ValidatePatientFields	Checks required patient fields and rejects invalid details.	PASS
3	TC-DOC-01	AddDoctor	Checks adding a doctor with valid details.	PASS
4	TC-DOC-02	RejectDuplicateDoctorId	Rejects a doctor record with an existing ID.	PASS
5	TC-DOC-03	ValidateDoctorFields	Checks required doctor fields and invalid details.	PASS
6	TC-DOC-04	SearchDoctors	Checks doctor search using supported criteria.	PASS
7	TC-PAT-04	SearchPatients	Checks patient search using supported criteria.	PASS
8	TC-APT-02	RejectUnknownPatient	Prevents booking for an unregistered patient.	PASS
9	TC-APT-01	BookAppointment	Checks booking with valid patient, doctor, date, and time.	PASS
10	TC-APT-04	ValidateAppointmentDateTime	Rejects invalid appointment date/time values.	PASS
11	TC-APT-03	RejectUnknownDoctor	Prevents booking with an unknown doctor.	PASS
12	TC-APT-06	CheckDoctorAvailability	Checks doctor availability before accepting an appointment.	PASS
13	TC-APT-05 / TC-NFR-08	PreventDoubleBooking	Prevents conflicting appointments for a doctor.	PASS
14	TC-APT-07	IgnoreCancelledAppointmentsInConflicts	Cancelled appointments do not block an available slot.	PASS
15	TC-APT-08	ViewAppointmentsDateTimeOrder	Checks appointment listing order by date and time.	PASS
16	TC-APT-09	CancelAppointment	Checks cancellation of a valid appointment.	PASS
17	TC-APT-10	RejectInvalidCancellation	Rejects invalid or unavailable cancellation requests.	PASS
18	TC-DOC-05	ViewAndListDoctors	Checks display and listing of doctor records.	PASS
19	TC-PER-01	StartupCSVLoading	Checks loading CSV records at startup.	PASS
20	TC-PER-02	MissingCSVFiles	Checks startup when expected CSV files are missing.	PASS
21	TC-PAT-05	PatientDetailsAndRegistrationPersistence	Checks persistence and retrieval of patient details.	PASS
22	TC-PER-03	SkipMalformedCSV	Checks safe handling of malformed CSV records.	PASS
23	TC-UI-01	MenuNavigation	Checks navigation through application menu options.	PASS
24	TC-UI-02	RejectInvalidMenuOptions	Checks safe rejection of invalid menu selections.	PASS
25	TC-PAT-06	UpdatePatientDetails	Checks updating permitted patient details.	PASS
26	TC-PAT-07	PreventPatientIdEdits	Prevents editing a patient identifier through the update workflow.	PASS
27	TC-DOC-06	UpdateDoctorDetails	Checks updating permitted doctor details.	PASS
28	TC-DOC-07	PreventDoctorIdEdits	Prevents editing a doctor identifier through the update workflow.	PASS
29	TC-PER-07	HandleCSVWriteFailures	Checks handling of CSV write failures and operation feedback.	PASS
30	TC-NFR-01	InvalidInputCrashResilience	Checks that invalid input does not cause an unhandled crash.	PASS
31	TC-PER-08	SkipDuplicateAndOrphanRecords	Checks handling of duplicate and orphaned persisted records.	PASS
32	CTest-only	OperationOutcomeMessages	Checks that operation outcomes are communicated.	PASS
33	TC-UI-04	ExitCleanly	Checks clean application exit through the normal workflow.	PASS
34	CTest-only	ExitProcessCode	Checks the process exit code for the tested shutdown path.	PASS
35	TC-NFR-02	RobustStartup	Checks startup under the tested startup conditions.	PASS
36	TC-NFR-03	DataRetention	Checks retention of required records across the tested lifecycle.	PASS
37	TC-NFR-05	UsabilityMenuDepth	Checks the intended menu-depth constraint.	PASS

No.	Test ID	Registered CTest name	Objective / behaviour checked	Result
38	TC-NFR-08	AppointmentConflictCorrectness	Checks correctness of appointment conflict detection.	PASS
39	TC-NFR-06	ArchitecturalClassIsolation	Checks separation between tested classes/modules.	PASS
40	TC-NFR-09	StandaloneOffline	Checks operation without network connectivity.	PASS
41	TC-NFR-04	Performance1000Records	Exercises the system with 1,000 records.	PASS
42	TC-SEC-01	AcceptListInputValidation	Checks validation of list-based input against accepted values.	PASS
43	TC-NFR-10	HumanReadableCSV	Checks that persisted CSV data remains human-readable.	PASS
44	TC-SEC-02	CSVFormulaInjectionGuard	Checks protection against spreadsheet formula injection in CSV data.	PASS
45	TC-SEC-03	OverLengthInputSafety	Checks safe handling of over-length input.	PASS
46	TC-SEC-06	RetainCancelledAppointmentRecords	Checks retention of appointment records after cancellation.	PASS
47	TC-SEC-07	MinimizePatientDataInAppointmentViews	Checks minimization of patient data in appointment views.	PASS
48	TC-SEC-09	SanitizeErrorMessages	Checks that error messages avoid unintended internal information disclosure.	PASS
49	TC-SEC-10	RestrictDataDirectory	Checks that data-file operations remain within the intended directory.	PASS
50	TC-APT-12	FilterUnknownEntity	Checks safe handling of unknown entity identifiers during appointment filtering.	PASS
51	TC-ID-01	DuplicateIdentityValidation	Checks duplicate identity validation in the tested registration path.	PASS

6. Assessment and limitations

The recorded run achieved a 100% pass rate: all 51 registered CTest tests passed, with no failures or skips. This demonstrates successful execution of the registered automated suite in the recorded environment. It does not establish that the application is defect-free or that all requirements and real-world scenarios have been validated.

The performance test is registered as a 1,000-record scenario; this report does not claim a specific latency or throughput value because no such measurement is included in the available execution summary. The test objectives are concise interpretations of registered names. Requirement-level traceability should be reviewed against the approved test-case document and SRS.

7. Evidence and reproducibility

Evidence artifact	Purpose
build-llvm/ctest-results.xml	JUnit-format record from the 51-test CTest run
build-llvm/production-coverage-final.txt	Production coverage summary
build-llvm/coverage.profdata	Merged LLVM instrumentation profile
Git revision 542b90be8adaddf13f7ce900340973eb4c9f5a56	Source revision associated with the recorded build

8. Conclusion

The Hospital Management System's registered automated suite completed with 51/51 tests passing, zero failures, zero skipped tests, and a recorded duration of 31.47 seconds. This report provides the test environment, functional-area summary, complete registered test inventory, and evidence references for the recorded run.

Note: the execution record should be retained with the exact source revision and generated JUnit XML when the reports are submitted or archived.